

BF528 Project 1

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1 Introduction

Alteromonas macleodii is a marine bacterium with a genome size of approximately 4.5 to 4.6 megabases[11]. The most recent reference assembly was generated using HGAP v2, achieving 98% accuracy, and was annotated with the NCBI Prokaryotic Genome Annotation Pipeline[1]. To replicate the current reference assembly and to gain greater familiarity with Nextflow, a custom pipeline was employed.

2 Materials & Methods

Short and long Nanopore sequencing reads for the *Alteromonas macleodii* genome were provided by Dr. Joseph Orofino. Quality filtering of long reads was performed with Filterlong v0.3.0[10], retaining reads with a minimum length of 500 bases and the top 90% ranked by quality. Filtered reads were compressed and stored in FASTQ format. Quality assessment of short reads was performed with FastQC v0.12.1[2].

Long reads were assembled using Flye v2.9.6 without a reference genome[4]. The assembly was indexed with Bowtie2 v2.5.4[5] and short reads were aligned to the indexed assembly using the same tool. Alignments were sorted with Samtools v1.22.1[6]. The draft assembly was then corrected and polished with Pilon v1.24[9], using the assembled genome as the `-genome` input and aligned short reads as the `-frags` input. The process was allocated 32 GB of memory using the `-Xmx32G` flag.

Assembly statistics and genome completeness were assessed with QUAST v5.3.0[3] and BUSCO v6.0.0[8]. The reference genome for *Alteromonas macleodii* was downloaded using the NCBI Datasets CLI v18.6.0[7] and compared to the initial unpolished assembly. QUAST was also used to compare the reference genome with the polished assembly from Pilon. BUSCO was run on the final polished assembly.

Finally, Prokka v1.14.6 was used to annotate the polished genome. The entire workflow was coordinated with Nextflow v25.04.6 and executed on the Massachusetts Green High Performance Computing Center, completing in approximately 1.5 hours.

3 Results

Using the described pipeline, the *Alteromonas macleodii* genome was successfully assembled. The assembly contained 4,627,920 base pairs with 5,830,642 reads. Of these, 5,687,099 reads were mapped. Fragment orientation was consistent, with FR orientation at 100%, and the mean insert size was 389 base pairs. Pilon reported 51x coverage and confirmed 99.9% of the original bases, correcting only 7 SNPs and one deletion. Several local continuity breaks associated with tandem repeats remained unresolved.

QUAST was used to assess assembly quality relative to the reference. The polished genome had a genome fraction of 98.429%, a duplication ratio of 1.001, 19 misassemblies, 18.18 misassemblies per 100 kbp, a total length of 4,627,920 base pairs, and a GC content of 44.7%. Compared to the unpolished assembly, the only difference was a slightly higher misassembly rate of 18.33 per 100 kbp. This indicates that polishing had little effect on large scale structural accuracy, and the Flye draft assembly was already of high quality.

BUSCO (Benchmarking Universal Single Copy Orthologs) was used to evaluate assembly completeness against a set of conserved orthologs from the OrthoDB database. BUSCO classifies matches as complete (C), duplicated (D), fragmented (F), or missing (M), reporting the total number of orthologs as n .

For the polished assembly, BUSCO reported C:99.6% [S:99.5%, D:0.1%, F:0.0%, M:0.4%, n:1828]. This indicates that nearly all expected orthologs were present, with only a very small fraction missing and almost no duplication. The absence of fragmented genes and the high per-

centage of single copy completeness support the conclusion that the assembly is both highly accurate and nearly complete.

Overall, the pipeline generated a high quality assembly. Across polishing, completeness assessment, and structural evaluation, only minimal errors were observed. Pilon confirmed 99.9% of bases, QUAST reported a genome fraction of 98.429%, and BUSCO reported 99.6% completeness. These results demonstrate that the assembly is accurate, complete, and structurally consistent.

4 Future Directions

Future work includes characterizing single nucleotide polymorphisms to investigate potential adaptive traits, identifying mobile genetic elements to study genome plasticity and evolution, and assessing functional homology with related organisms to determine conserved and unique biological pathways.

Variant calling pipelines can predict the functional effects of SNPs, which may be validated experimentally. Assays measuring nitrogen production, heat tolerance, antibiotic resistance, and surface adhesion could help determine whether specific SNPs confer a biological advantage.

Mobile genetic elements can be identified with detection pipelines that locate both original and new insertion sites. If genes are disrupted, their impact on biology can be assessed.

Conserved sequences can be identified using BLAST, and metabolic pathways can be characterized to evaluate whether *Alteromonas macleodii* exhibits capacities similar to or different from related species.

Conclusions

The custom pipeline successfully assembled the *Alteromonas macleodii* genome with high accuracy and minimal polishing. Tools including FastQC, Flye, Filtlong, Samtools, Bowtie2, and Pilon performed effectively. Genome completeness and quality were confirmed with BUSCO and QUAST.

It is important to note that the quality of the assembly depends on the input reads. High quality reads were provided for this project, which contributed to the final results. Users of this pipeline may need to adjust filters in Flye and FastQC, account for misalignments, and check for sample contamination.

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References

- [1] *Alteromonas macleodii* genome assembly ASM284987v1.
- [2] Babraham Bioinformatics - FastQC A Quality Control tool for High Throughput Sequence Data.
- [3] Alexey Gurevich, Vladislav Saveliev, Nikolay Vyahhi, and Glenn Tesler. QUAST: quality assessment tool for genome assemblies. *Bioinformatics*, 29(8):1072–1075, April 2013.
- [4] Mikhail Kolmogorov, Jeffrey Yuan, Yu Lin, and Pavel A. Pevzner. Assembly of long, error-prone reads using repeat graphs. *Nature Biotechnology*, 37(5):540–546, May 2019. Publisher: Nature Publishing Group.
- [5] Ben Langmead and Steven L Salzberg. Fast gapped-read alignment with Bowtie 2. *Nature methods*, 9(4):357–359, March 2012.
- [6] Heng Li, Bob Handsaker, Alec Wysoker, Tim Fennell, Jue Ruan, Nils Homer, Gabor Marth, Goncalo Abecasis, and Richard Durbin. The Sequence Alignment/Map format and SAMtools. *Bioinformatics*, 25(16):2078–2079, August 2009.
- [7] Naomi A. O’Leary, Eric Cox, Jennifer B. Holmes, William R. Anderson, Robert Falk, Valerie Hem, Mark T. N. Tsuchiya, Gregory D. Schuler, Xiaojian Zhang, James Torcivia, Andrew Ketter, Laura Breen, Joshua Cothran, Hammad Bajwa, James Tinne, Patricia A. Meric, William Hlavina, and Valerie A. Schneider. Exploring and retrieving sequence and metadata for species across the tree of life with NCBI datasets. *Scientific Data*, 11(1):732, July 2024.
- [8] Fredrik Tegenfeldt, Dmitry Kuznetsov, Mosè Manni, Matthew Berkeley, Evgeny M Zdobnov, and Evgenia V Kriventseva. OrthoDB and BUSCO update: annotation of orthologs with wider sampling of genomes. *Nucleic Acids Research*, 53(D1):D516–D522, January 2025.
- [9] Bruce J. Walker, Thomas Abeel, Terrence Shea, Margaret Priest, Amr Abouel-liel, Sharadha Sakthikumar, Christina A.

- Cuomo, Qiandong Zeng, Jennifer Wortman, Sarah K. Young, and Ashlee M. Earl. Pilon: An Integrated Tool for Comprehensive Microbial Variant Detection and Genome Assembly Improvement. *PLOS ONE*, 9(11):e112963, November 2014. Publisher: Public Library of Science.
- [10] Ryan Wick. `rrwick/filtlong`, September 2025. original-date: 2017-07-25T04:22:25Z.
- [11] Matthias Wietz, Mario López-Pérez, Daniel Sher, Steven J. Biller, and Francisco Rodriguez-Valera. Microbe Profile: *Alteromonas macleodii* – a widespread, fast-responding, ‘interactive’ marine bacterium. *Microbiology*, 168(11):001236, 2022. Publisher: Microbiology Society,.